Week 11

Introduction to Architectural Acoustics

Spring 2021
Kurt Neiswender, AIA
College of Architecture and Design
Lawrence Technological University

IMPORTANCE OF ACOUSTICS

- We experience spaces aurally as well as visually
- Appropriate acoustics within a space is every bit a part of **occupant comfort** and **room functionality** as other factors

Examples:

- Hospitals: better healing with fewer sleep disturbances
- Schools: better learning and comprehension when every word is heard
- Open plan offices: fewer work disturbances with appropriate background sound levels
- Councilor, legal, medical offices & rooms: private conversations can remain private (people withhold information when they feel their privacy is at risk)

MAIN TOPICS in Acoustics

- Fundamentals of sound and measurement
- Barriers and absorbers
- Sound isolation
- Noise control in buildings
- Behavior of sound in rooms
- Mechanical noise control
- Speech intelligibility and privacy
- Applications in spaces
    - (e.g., lecture halls, cinemas, conference rooms, etc.)

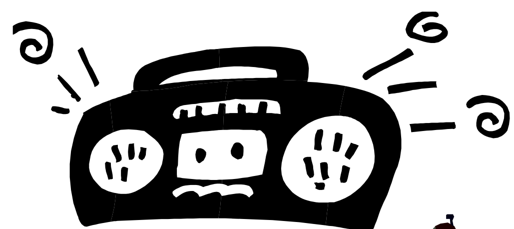
SOUND & NOISE

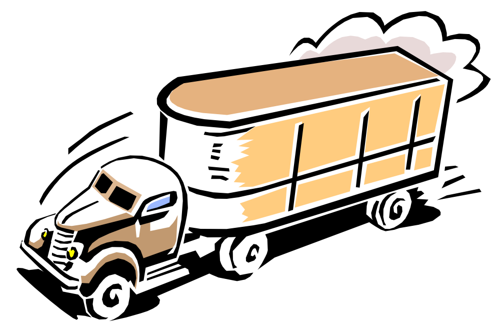
“***Sound*** is a vibration in an elastic medium such as air, water, most building materials, and the earth.”

--M. David Egan, Architectural Acoustics, © 2007, p. 2.

Noise is unwanted sound

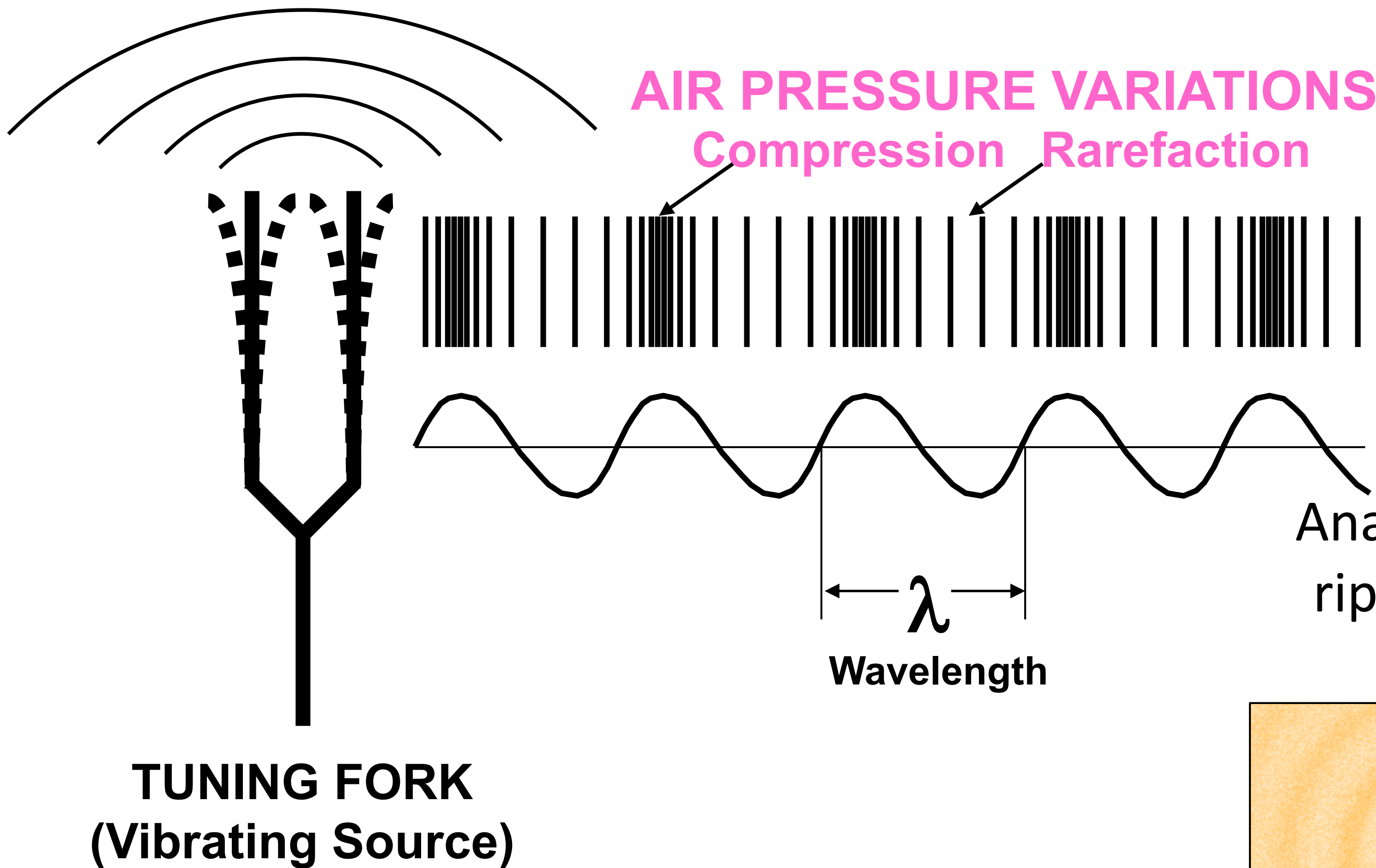
– “Noise” is derived from the word “nausea”



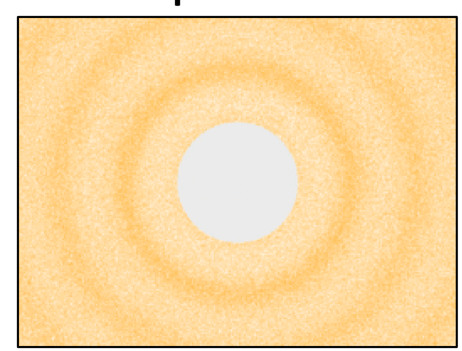


HOW SOUND IS CREATED

- Vibrating air particles convey sound waves

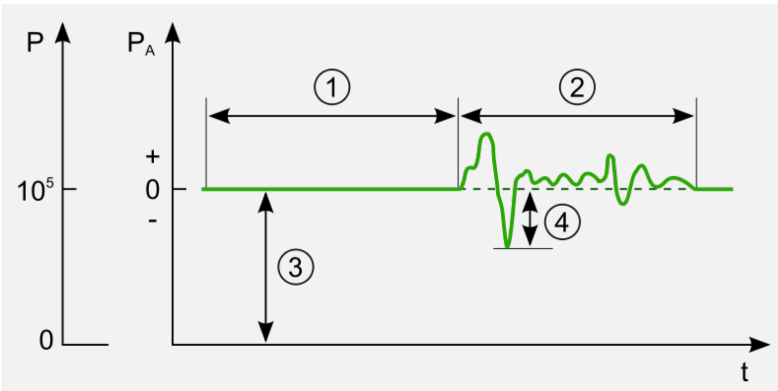


Analogous to ripples on a pond



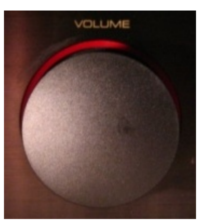
TIME AND FREQUENCY

- Sound at a specific position

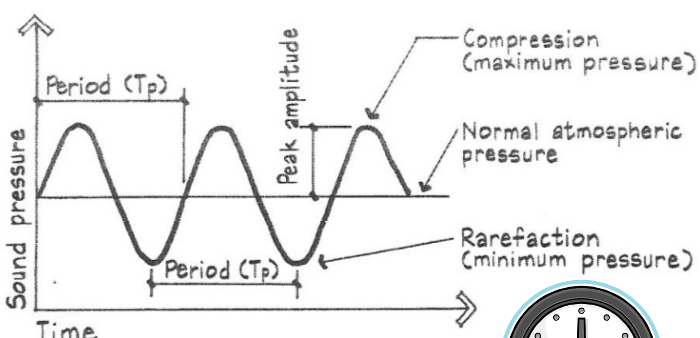


Sound pressure diagram:
1. silence
2. audible sound
3. atmospheric pressure
4. instantaneous sound pressure

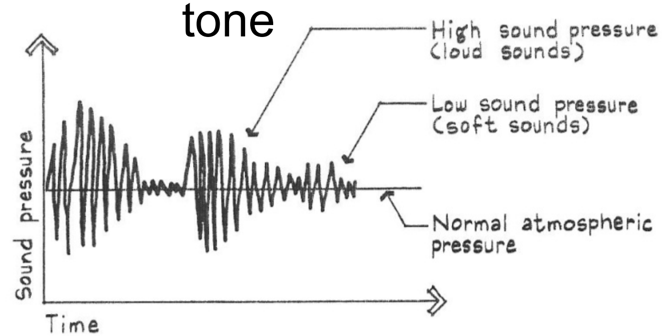
- We represent this movement on a time graph as:



Pure tone

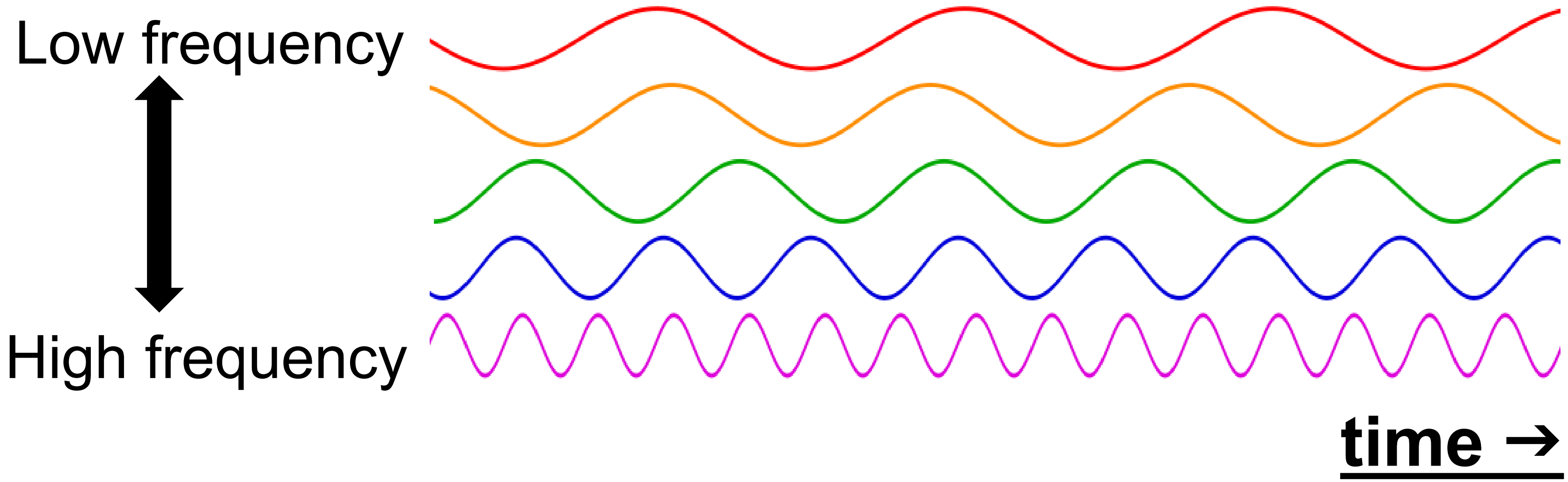


Complex tone

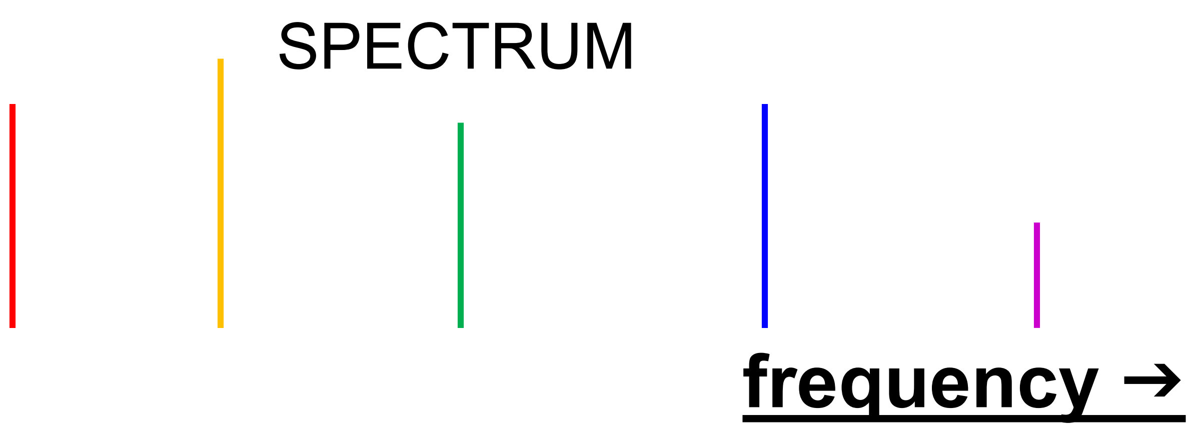


TIME AND FREQUENCY

- Complex sounds can be represented as a set of discreet frequencies



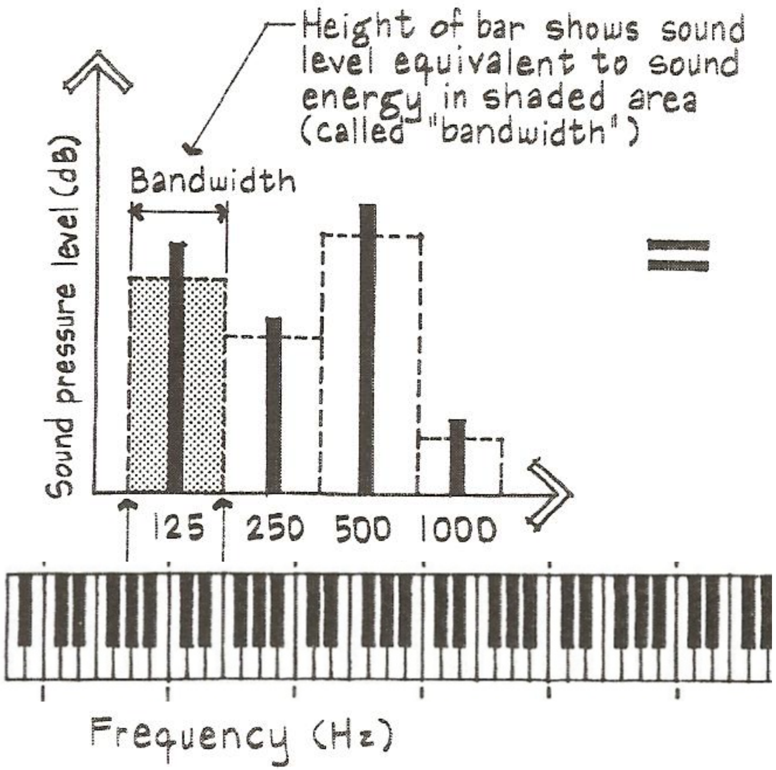
- Discreet frequencies are shown collectively into a noise spectrum



FREQUENCY

- **Frequency**, f, is the rate of repetition of a periodic event.
- ***The number of repeated cycles in one second*** is the frequency of the sound in the units of hertz (Hz) and kilohertz (kHz). 1000 Hz = 1 kHz
- Occasionally referred to as "cycles per second"

FREQUENCY



- **A frequency band** is a range of frequencies identified by their center frequency
- Many frequencies occurring at once is called **broadband**
- An octave is a <u>doubling in</u> frequency.
  - ex. 125 Hz and 250 Hz
- A sound spectrum (a frequency graph) describes the magnitude (volume) of sound energy at many frequencies

FREQUENCY BANDS

- Frequency bands have an upper and lower cutoff (limit) frequency and are identified by their center frequency
- In architectural acoustics, octave and 1/3-octave bands are most typically used
  - Note the logarithmic scale which represents human hearing

TABLE 4.2

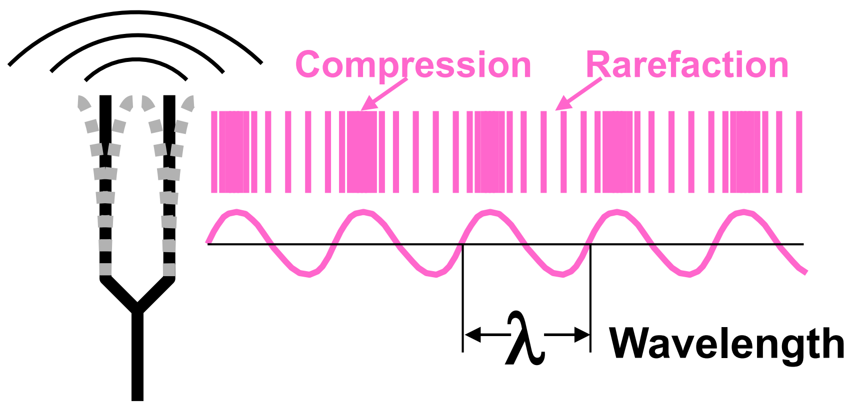
Comparison of 1-octave and $\frac{1}{3}$-octave bands

1 Octave			$\frac{1}{3}$ Octave		
Lower cutoff frequency (Hz)	*Center frequency (Hz)*	*Upper cutoff frequency (Hz)*	*Lower cutoff frequency (Hz)*	*Center frequency (Hz)*	*Upper cutoff frequency (Hz)*
11	16	22	14.1	16	17.8
			17.8	20	22.4
			22.4	25	28.2
22	31.5	44	28.2	31.5	35.5
			35.5	40	44.7
			44.7	50	56.2
44	63	88	56.2	63	70.8
			70.8	80	89.1
			89.1	100	112
88	125	177	112	125	141
			141	160	178
			178	200	224
177	250	355	224	250	282
			282	315	355
			355	400	447
355	500	710	447	500	562
			562	630	708
			708	800	891
710	1,000	1,420	891	1,000	1,122
			1,122	1,250	1,413
			1,413	1,600	1,778
1,420	2,000	2,840	1,778	2,000	2,239
			2,239	2,500	2,818
			2,818	3,150	3,548
2,840	4,000	5,680	3,548	4,000	4,467
			4,467	5,000	5,623
			5,623	6,300	7,079
5,680	8,000	11,360	7,079	8,000	8,913
			8,913	10,000	11,220
			11,220	12,220	14,130
11,360	16,000	22,720	14,130	16,000	17,780
			17,780	20,000	22,390

Irwin & Graf, *Industrial Noise and Vibration Control,* p. 93, 1979

TIME, FREQUENCY & WAVELENGTH

Wavelength, λ, is the physical distance a sound wave travels during one cycle of vibration



$$\lambda = \frac{c}{f}$$ **general equation**

λ = wavelength (ft)
f = frequency (Hz)
c = speed of the wave

speed of the sound in air:
1130 ft/s (344 m/s)
Therefore:

$$\lambda = \frac{1130}{f}$$ **for air**

Frequency (Hz)	Wavelength in Air (mm)	(inches)
125	2744	108.3
400	858	33.8
1000	343	13.5
5000	69	2.7
8000	43	1.7

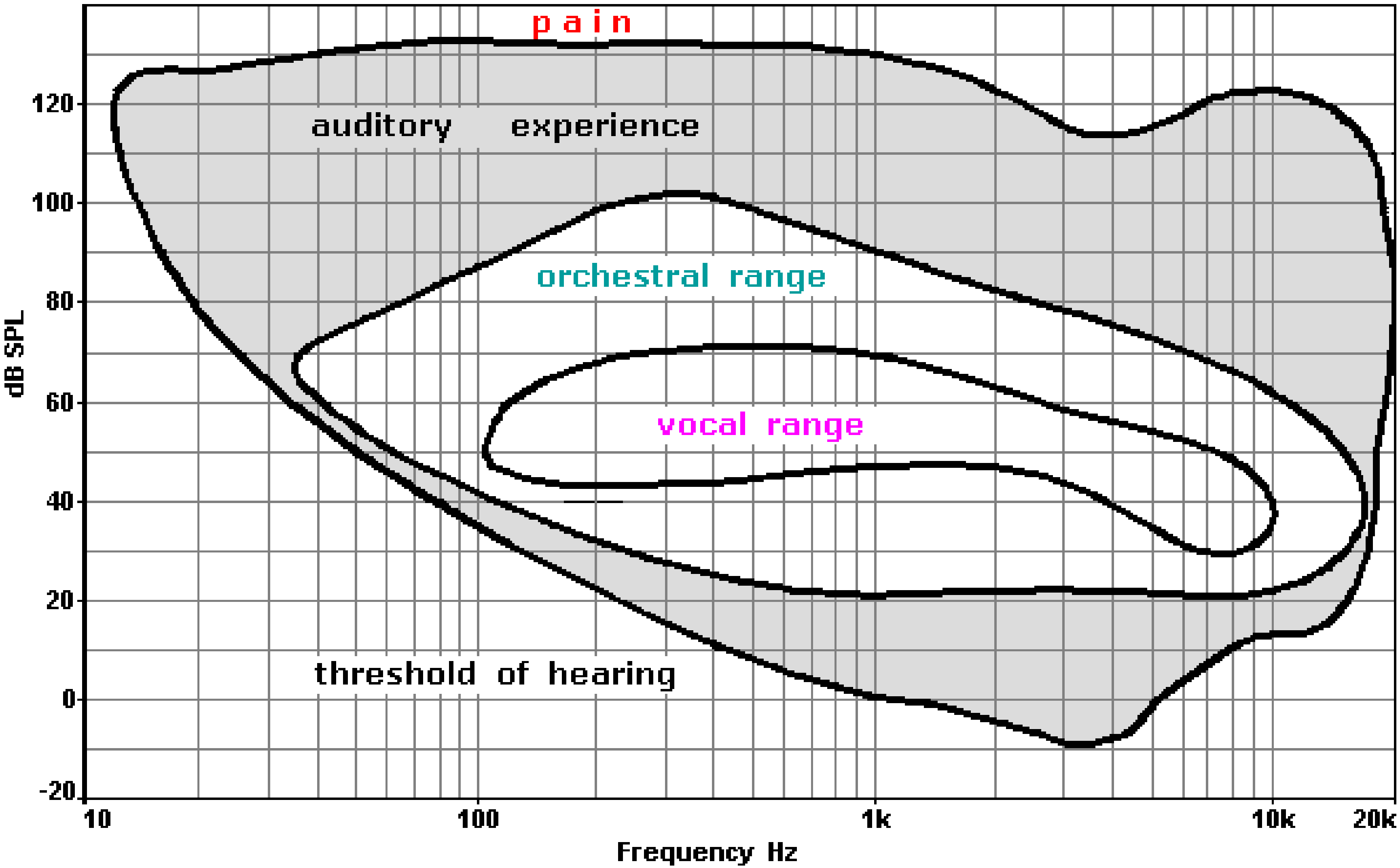
$$T_p = \frac{1}{f}$$

A mathematical relationship for period and frequency

ACOUSTIC FREQUENCIES

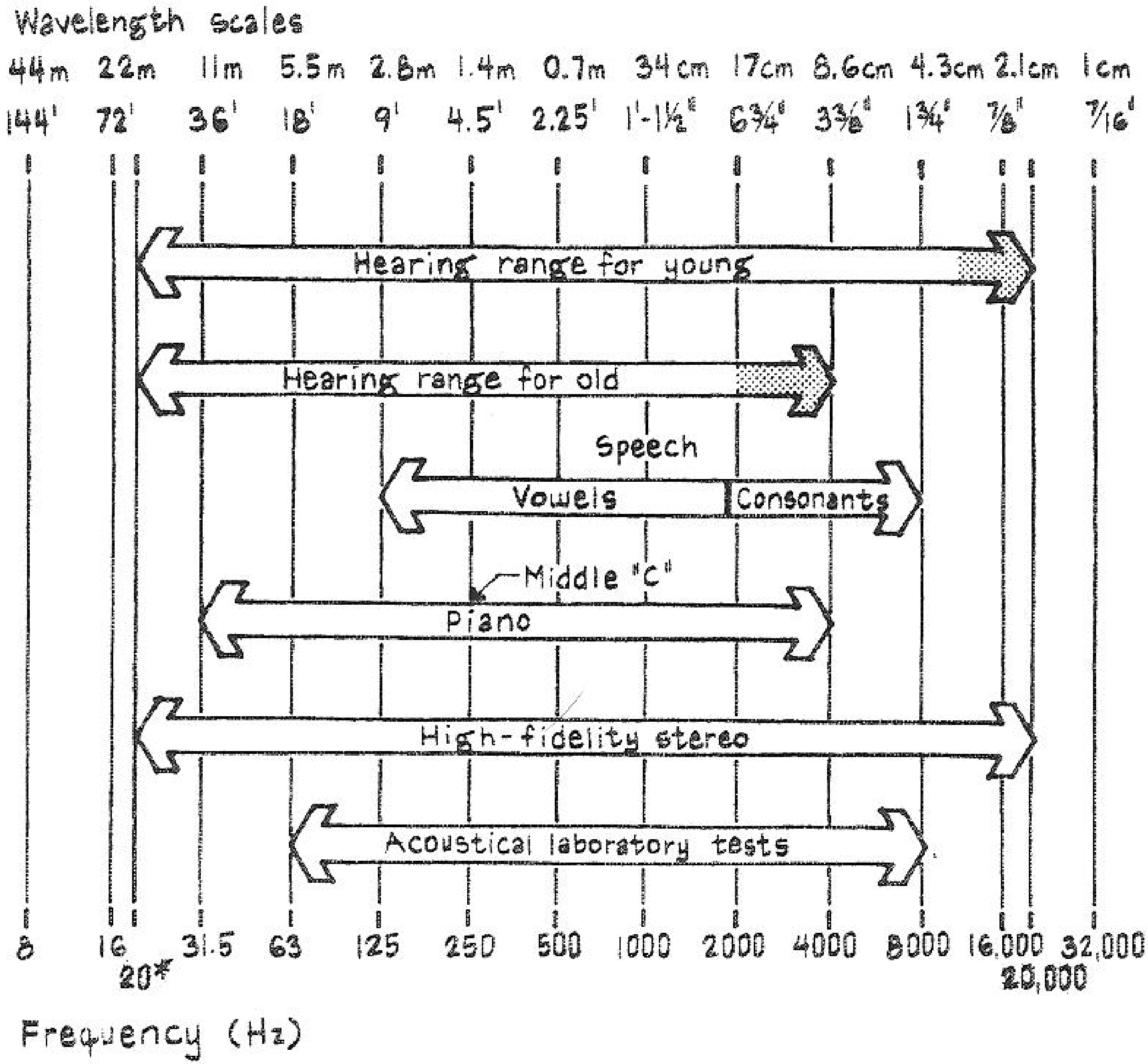
- For convenience, it is said that audible frequencies to humans are between **20 Hz and 20 kHz**
- Hearing may not be quite that good when young and high frequency hearing deteriorates with age and/or hearing damage
- Above the human hearing range is the **ultrasonic** region
- Below is called the **infrasonic** range

ACOUSTIC FREQUENCIES



Egan, Architectural Acoustics

ACOUSTIC FREQUENCIES



*Vibrations below 20 Hz are *not* audible, but can be felt.

Egan, Architectural Acoustics

SOUND AMPLITUTE

- Almost all hearing sensations are proportional to the intensity of the stimulus
- The intensity range that humans can hear over is enormous (a ratio of 10 trillion from the quietest to loudest sound)
- It is convenient to collapse this scale to a workable range by converting to decibels (dB).

$$L_I = 10 \log \frac{I}{I_o}$$

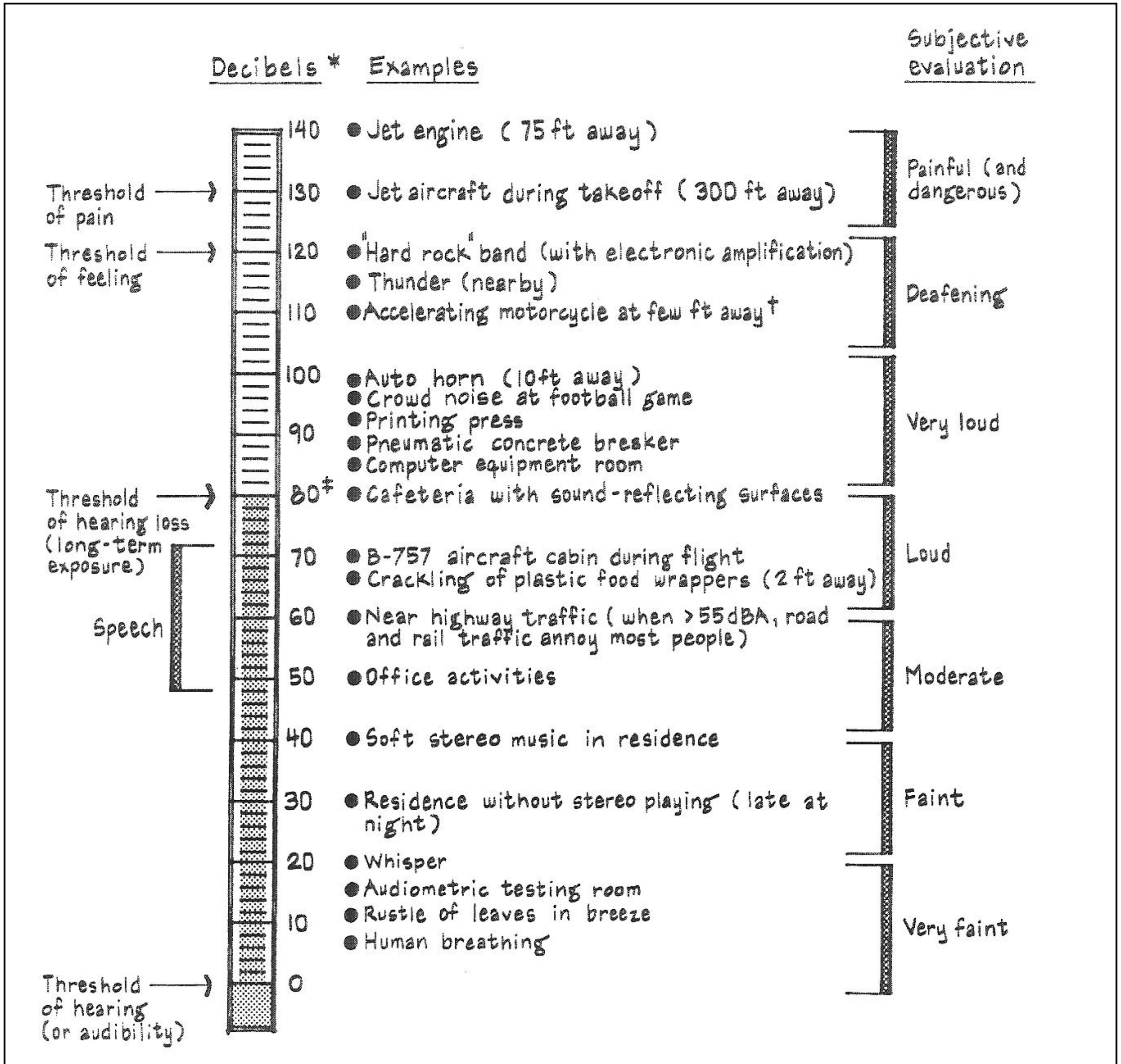
where

L_I = sound intensity level

I = sound intensity (W/m^2)

I_o = reference sound intensity, 10^{-12} (W/m^2)

SOUND AMPLITUTE



Egan, Architectural Acoustics

SOUND INTENSITY AND SOUND PRESSURE

- Sound intensity is proportional to the square of the sound pressure

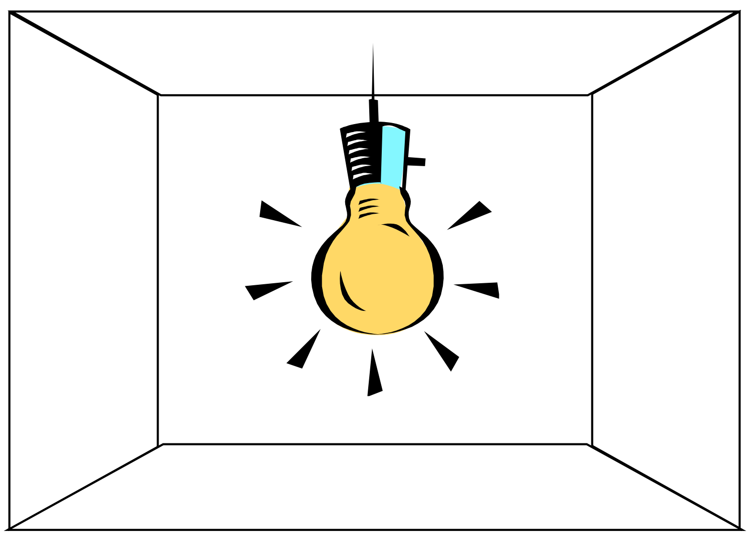
$$I \propto p^2$$

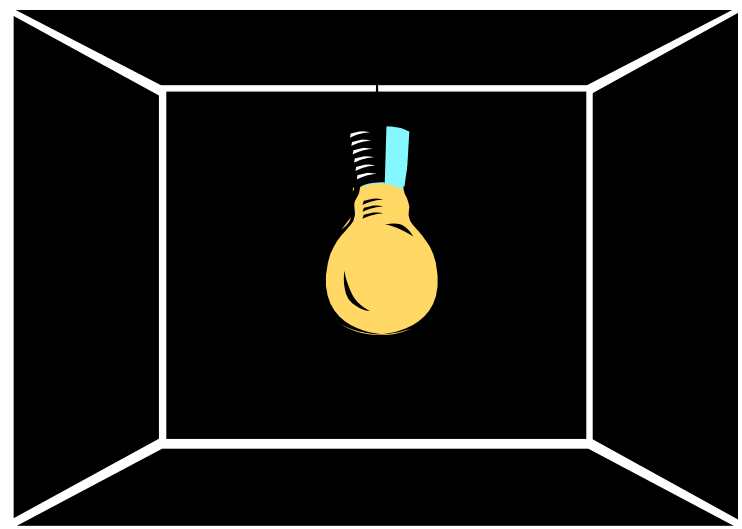
- Like sound intensity and sound intensity level, sound pressure can be expressed in terms of a sound pressure level in dB.
- We can measure sound pressure levels (L_p or SPL) with sound level meters very easily
- **In architectural acoustics, L_p is approximately equal to L_I.** We will use L_p for our discussions and calculations

SOUND POWER vs SOUND PRESSURE

Sound Power ⟷ Electrical Power

Sound Pressure ⟷ Illumination





100 W bulb ⟵ Power is same ⟶ 100 W bulb

bright room ⟵ Intensity varies With surroundings ⟶ dark room

SOUND POWER vs SOUND PRESSURE

SOUND POWER, L_W	SOUND PRESSURE, L_P
Characteristic of **sound sources** such as fans & chillers	Characteristic of a sound source **in a specific environment**
Independent of location, direction of sound and environment (i.e., constant for any environment)	**Dependent** on location, direction of sound & environment (should include distance and conditions for the measurement)
Used in analysis	**What is sensed by the eardrum**
Determined analytically in acoustics laboratories; Cannot be measured directly	Can be **measured** easily with a microphone
Direct product comparison possible	Product comparison valid only for same test method

DECIBEL SCALE – L_W

Sound power level (L_W or PWL) uses 10^{-12} watts (W) as its reference quantity and is expressed in dB also

$$L_W = PWL = 10\log\left(\frac{W}{W_o}\right)$$

Notice that after the conversion from a “raw” quantity to a decibel quantity, the word “level” is appended to the quantity (i.e., sound power and sound power level)

DECIBEL SCALE – L_P

Sound pressure level (L_p or SPL) is different because it is a squared quantity. The reference quantity is 20×10^{-6} pascals (Pa), also written 20 µPa. dB is the unit for SPL.

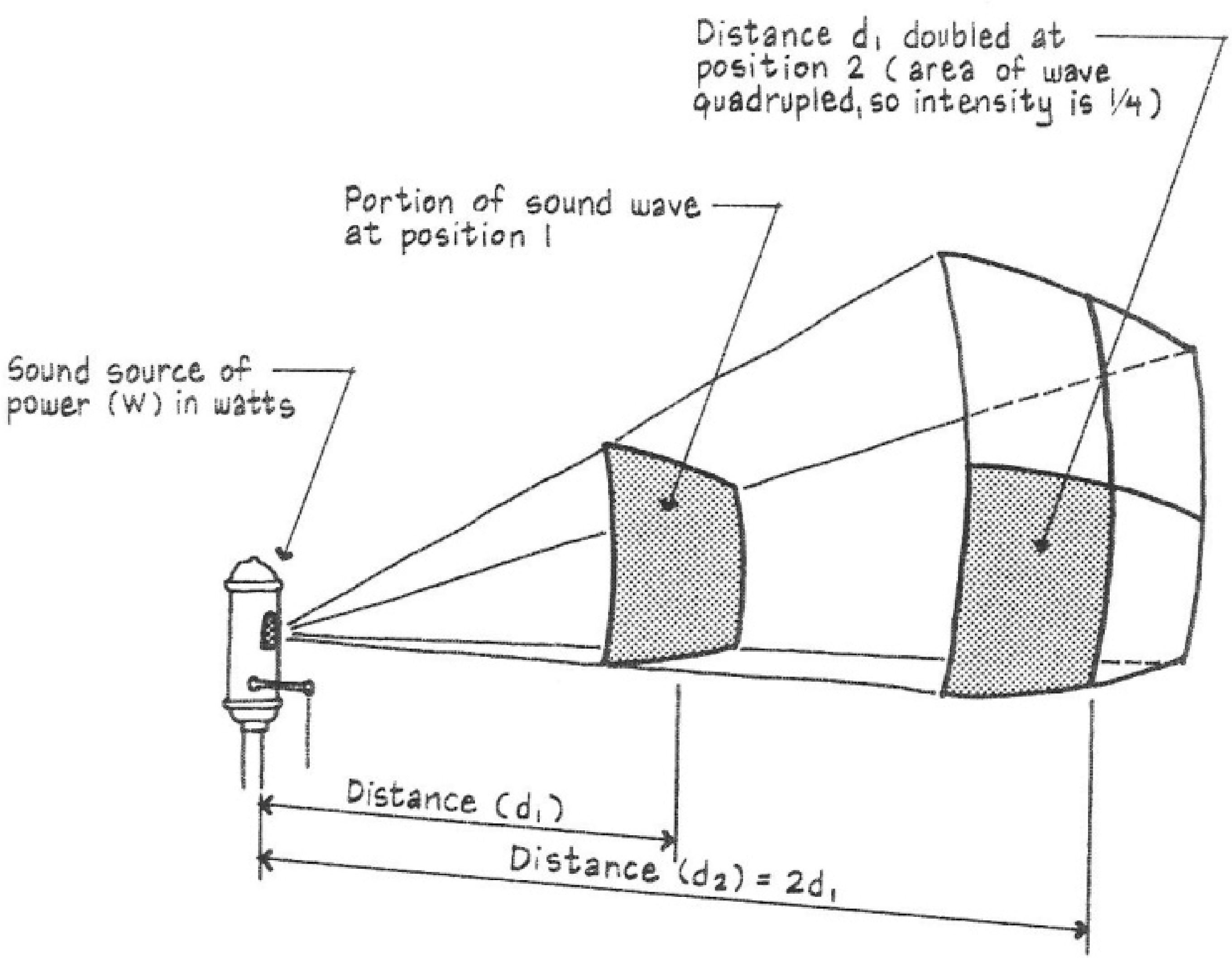
$$L_p = SPL = 10\log\left(\frac{p^2}{p_o^2}\right)$$

$$L_p = SPL = 10\log\left(\frac{p}{p_o}\right)^2$$

$$\boxed{L_p = SPL = 20\log\left(\frac{p}{p_o}\right)}$$

SOUND PRESSURE VS DISTANCE

SOUND ATTENUATION BY DISTANCE FOR A POINT SOURCE



SOUND PRESSURE VS DISTANCE

SOUND ATTENUATION BY DISTANCE FOR A POINT SOURCE

Sound generally radiates spherically from a source after the observer is several wavelengths away from the source. It is behaving as a point source in this region.

The sound intensity is inversely proportional to the square of the distance in a free field.

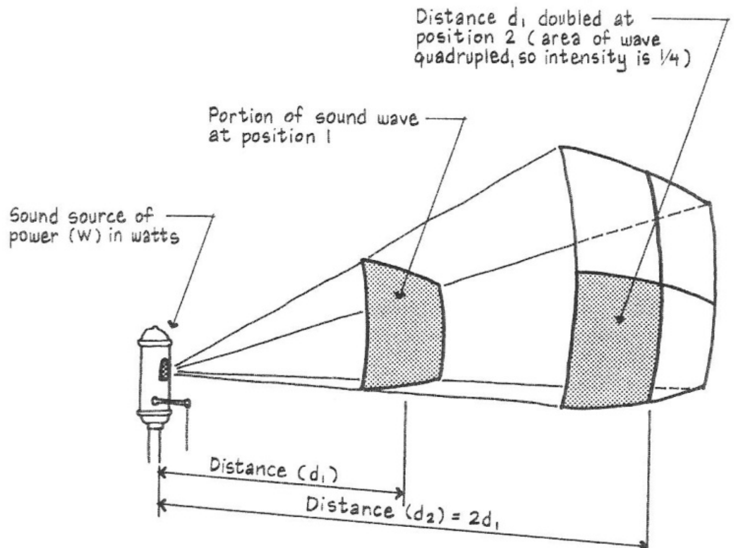
$$I = \frac{W}{4\pi d^2}$$

where:

W = power in watts

d = distance from source

$4\pi d^2$ = surface area of a sphere



Egan, Architectural Acoustics

SOUND PRESSURE VS DISTANCE

SOUND ATTENUATION BY DISTANCE FOR A POINT SOURCE

This relationship is called the inverse square law and is only valid for free fields (i.e., open space with no reflections).

We can use the inverse square law to predict an SPL at a specified distance from a source at a different distance using the following equation:

$$L_{p2} = L_{p1} - 20\log\left(\frac{d_2}{d_1}\right)$$

The difference between two SPLs is called noise reduction.

$$NR = L_{p2} - L_{p1} = 20\log\left(\frac{d_2}{d_1}\right)$$

SOUND PRESSURE VS DISTANCE

SOUND ATTENUATION BY DISTANCE FOR A POINT SOURCE

- When doubling the distance, the sound pressure level **decreases by 6 dB**
- When halving the distance, the sound pressure level **increases by 6 dB**

$$SPL_2 = SPL_1 - 20\log\left(\frac{d_2}{d_1}\right)$$

$$\text{if } d_2 = 2d_1$$

$$SPL_2 = SPL_1 - 20\log\left(\frac{2d_1}{d_1}\right)$$

$$SPL_2 = SPL_1 - 20\log(2)$$

$$\boxed{SPL_2 = SPL_1 - 6 \qquad (\text{if } d_2 = 2d_1)}$$

ADDING SOUND PRESSURE LEVELS

- Since sound levels are in terms of logarithms, they must be added and subtracted logarithmically.

Combine intensities (where $I \propto p^2$)

$$Lp_n = 10\log[\frac{p_1{}^2 + p_2{}^2 + \ldots + p_n{}^2}{p_{ref}{}^2}]$$

$$Lp_n = 10\log[\frac{p_1{}^2}{p_{ref}{}^2} + \frac{p_2{}^2}{p_{ref}{}^2} + \ldots + \frac{p_n{}^2}{p_{ref}{}^2}]$$

since $L_p = 10Log(\frac{p_1{}^2}{p_{ref}{}^2})$ $\Rightarrow$ $\frac{p_1{}^2}{p_{ref}{}^2} = 10^{(L_p/10)}$

therefore $Lp_n = 10\log[10^{(Lp_1/10)} + 10^{(Lp_2/10)} + \ldots + 10^{(Lp_n/10)}]$

EXAMPLES FOR ADDING SOUND LEVELS

Find the sound level when the following are added:

(a) 51 dB and 51 dB $Lp_n = 10\log[10^{(51/10)} + 10^{(51/10)}] = 54dB$

(b) 90 dB and 50 dB $Lp_n = 10\log[10^{(90/10)} + 10^{(50/10)}] = 90dB$

(c) 45 dBA, 42 dBA and 40 dBA

$$Lp_n = 10\log[10^{(45/10)} + 10^{(42/10)} + 10^{(47/10)}] = 50dBA$$

SUBTRACTING SOUND LEVELS

$$Lp_n = 10\log[10^{(Lp_1/10)} - 10^{(Lp_2/10)} - \ldots - 10^{(Lp_n/10)}]$$

Practice 1

What is the effect of turning off two machines, one at 83 dB and the other at 86 dB, when the overall sound level in a workshop is 90 dB?

Answer

$$Lp_n = 10\log[10^{(90/10)} - 10^{(83/10)} - 10^{(86/10)}]$$

$$Lp_n = 86dB$$

ADD AND SUBTRACT SOUND LEVELS

Practice 2

The maximum allowable level in a space is 60 dBA. The existing sources produce 53 dBA. What is the noisiest machine that can be added to the space?

Answer

$$Lp_2 = 10\log[10^{(60/10)} - 10^{(53/10)}] = 59dB$$

Week 10

Sound Measurement + Hearing

Spring 2021
Kurt Neiswender, AIA
College of Architecture and Design
Lawrence Technological University

SOUND LEVEL METERS

Sound pressure is measured with a Sound Level Meter (SLM). Meters typically have these features:

- Microphone
- Digital or analog needle readout of level
- Time averaging speeds (Slow, Fast, Impulse, Peak)
- Weighting networks (A, B and C weighting)
- Ability to measure sound in frequencies of 20 Hz to 20 kHz
- Ability to filter frequencies into octave or 1/3 octave bands

SLM EXAMPLES

www.radioshack.com

SOUND LEVEL METERS

Time Averaging Speeds

- SLOW – sound level averaged over 1 sec
  used with steady state sources or slowly changing noise levels

- FAST – sound level averaged over 1/8 second (125 milliseconds
  used when a sound varies with time faster than 1 second

- IMPULSE – sound level averaged over 35 milliseconds when sound rises and 1500 ms when sound decreases
  used for brief rapidly rising sounds like press machine

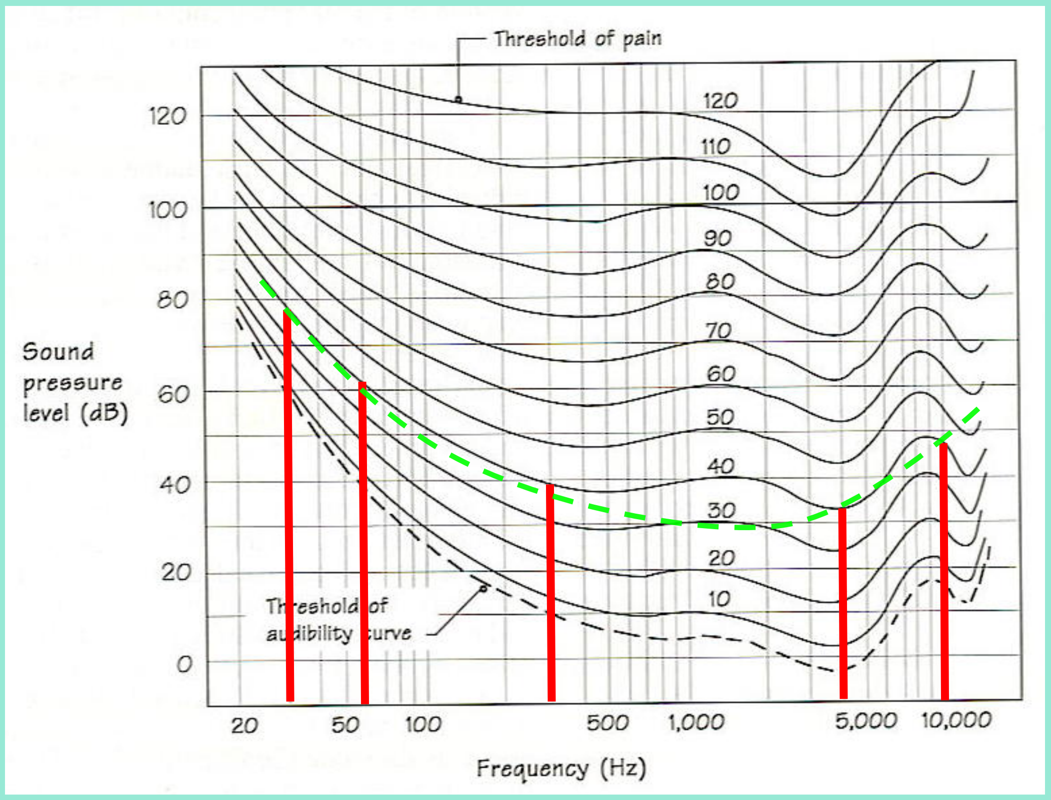
- PEAK – sound level averaged over 0.01 seconds or less
  used to monitor extremely short sounds like a gun shot

EQUAL LOUDNESS AND WEIGHTING

- Recall that humans hear roughly from 20 Hz to 20 kHz. (Bats can hear higher frequencies. Whales can hear lower frequencies.)
- We do not hear SPLs equally over that range. In response to this, researchers created an equal loudness contours plot from many subjective tests

EQUAL LOUDNESS AND WEIGHTING

Interpretation: Tones of different dB levels are perceived to be the same loudness (in phon) if they touch the same equal loudness curve.



All these tones (in red) are 40 phon and thus "equally loud"

78 dB @ 30 Hz
62 dB @ 60 Hz
39 dB @ 300 Hz
33 dB @ 4 kHz
47 dB @ 10 kHz

CHANGES IN LEVELS

Our perception of noise level is not linear – follows closely to the logarithmic scale

Change in L_p (dB)	Human Perception
1	Imperceptible
3	Just Perceptible
5	Clearly Noticeable
10	Substantial Change (Twice as much)
20	Four Times as Loud

CHANGES IN LEVELS

General "Rules of Thumb"

- **Decibel arithmetic works on a log scale.**
- **A 3 dB reduction in SPL requires a 50 % reduction in sound power in the noise.**
- **A 5 dB reduction in SPL requires a 70 % reduction in sound power in the noise.**
- **A 10 dB reduction in SPL requires that the sound power be reduced by a factor of 10 !**

→ ***Our ears will perceive this as half as loud.***

Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Acoustics 1 Lecture slides, College of Architecture + Planning, University of Utah
- www.archdaily.com
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4